A toy malign-prior question with an answer: bounded Solomonoff prediction in a world of programs

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Abstract

The "universal prior is malign" argument, in the form we examine, says that where the predictions of a Solomonoff inductor are actually used in a critical way in some universe, a lot of what happens is malign: takeovers through the prediction channel. We give a toy version in which this is a mathematical statement and work out what can be proved there. Universes are programs for a fixed machine U. Some of them make a *critical use* of a fixed Solomonoff predictor, possibly time- and length-bounded: they hand it their output so far and run the continuation they get back as a program. A hypothesis *reproduces itself* if the continuation it supplies is its own code; the world then runs that hypothesis. Let H be the prior probability of a critical use and κ the share of the posterior, at the context of a use, of the worlds that make it. Two questions have to be kept apart.

- **Reach of one seeker.** Without resource bounds, for each continuation there is a program, of some length a, that enumerates the programs making a critical use, predicts their output up to the use, and then supplies that continuation. It has posterior share more than $2^{-a-1}\kappa/H$ at every use, and its continuation is handed over in worlds of total prior probability more than 2^{-a-1} times the average of κ, and at most H. Once uses are rarer than half the seeker's weight, its programs outweigh the users at their contexts. Supplying its own code needs a longer program and a user whose program is short enough for the block.
- **Fate of the uses.** This does not show that most uses end in a takeover. The share that do is the posterior share of such programs at the contexts of uses. A seeker that searches has an honest twin, which makes the same search and then continues as the user it found; the twin is in the posterior with 2^{a-h} times the seeker's mass, h being its length. So where the users go on to print the block themselves, that seeker's programs supply what the predictor returns with probability at most $1/(1+2^{a-h})$. The bound is for one seeker and one search at a time, and the masses of different seekers add.
- **No goal has to be written down.** Any program of length a whose output puts on each context of a class of uses at least half that context's share of the class, for instance a simple world with a cheap readout whose inhabitants choose to do this, outweighs the worlds that make those uses as soon as the class has total prior mass below 2^{-a-1} and the program's own outputs are not counted as uses. The twin bound does not apply to a program that is not a written search followed by a continuation. A time bound does where, as in this example, the program prints nothing until a long computation has run: a schedule shorter than that start-up gives it no mass.
- **Without search.** At every use that a fixed test recognises, a wrapper around the explanations of the context gives any fixed continuation at least a share 2^{-c}, with c the wrapper's length, whatever the world and the amount of data; under resource bounds this is multiplied by the share of explanations with room to spare.
- **Calibration and budget.** If worlds are drawn from the predictor's own prior and go on within its bounds, the continuation the predictor samples and the world's own continuation have the same distribution given the context. Along any world the predictor can express, the posterior mass off the true continuation, summed over all cut points, is at most the block length times the world's description length.

- **Other ways to a majority**, each an inequality: a machine built for it (any share can be arranged by choosing the machine); a cheaper explanation of the context that is not a use, for example a copy of the data stream at a place in the world that is simpler to specify than the sensor; or a world too expensive for the predictor to simulate.
- **Feedback.** If the world does what is predicted, a sampled self-fulfilling deviation removes the default hypotheses for good, and a time-bounded predictor confines such a world to futures that are cheap to compute.

An exact computation on a small machine shows sizes: with programs of at most ten commands, most of the posterior leaves one stream's true continuation at bit 17, because three-command programs explain its first 16 bits; the effect weakens as longer programs are admitted.

1 The question

Solomonoff induction predicts a sequence by weighting every program that could have produced it by $2^{-\text{length}}$. Christiano [1] argued that this prior is *malign* (see also [2]). We take the claim in the following form: *where the predictions of such an inductor are actually used in a critical way in some universe, a lot of what happens is malign, namely takeovers through the prediction channel.* The mechanism offered is that some short programs simulate worlds whose inhabitants want to influence whoever uses the prior; they look for places where a prediction is about to be used, output accurate predictions to gain posterior weight there, and then output something that serves them. The argument is informal at two places. It needs a claim about which programs are short on a "natural" machine, and it needs a notion of what it is for a prediction to be used. (We could not open the two posts while writing; our account of the argument is from memory and from summaries by others.)

This note asks what remains when both places are filled in by a toy. The toy follows two suggestions of Kaarel Hänni. In the first, a computable world (a cellular automaton, say) contains a sensor, a large computer that runs a time-bounded Solomonoff induction on the sensor's data, and an actuator driven by the predictions; one asks whether the predictor ends up in control, either over a long run or at once. In the second, which we take as the main definition, *worlds are just programs*: some of them hand their output so far to a fixed bounded Solomonoff predictor and run the continuation they receive, and one asks whether hypotheses that "look for such opportunities, predict well up to them, and then output themselves" get a decent share. Section 2 gives the definitions. The first suggestion becomes the special case in which the context handed to the predictor is read off at one location of one big world (Sections 4.2 and 5).

Read in the toy, the claim splits into two questions that should be kept apart. *Reach:* how far does one program get that looks for critical uses, predicts well up to them, and then supplies a continuation of its choice? *Fate of the uses:* what share of the critical uses end in a takeover? The first has a clean answer without resource bounds: for each continuation there is such a program whose continuation is handed over in worlds of total prior probability more than half its own weight times the average share of the users among the programs that print their contexts. That does not answer the second, which is the one the claim is about. For it the toy gives a lower bound, by adding up seekers, and an upper bound on the part of one seeker that finds its uses by search: the same search followed by continuing as the user that was found is an honest hypothesis with a mass that differs from the seeker's by a factor comparing the lengths of the two programs (Theorem 13 and Proposition 14). That bound does not cover a seeker whose search and whose goal are not written in its text, such as a simple world whose inhabitants choose to do this; for such a program the toy has the reach statement, with the length of the world in place of the length of an explicit seeker, and a time bound as the only thing shown to exclude it.

Results. Unless said otherwise, statements are for a fixed monotone machine U and the mixture ξ of its programs of length at most L that keep to a time schedule T (Definition 1).

1. *Calibration* (Theorem 9). If the world is drawn from the same bounded prior and goes on to print the next block within the bounds, then given the context the predictor's sampled continuation and the world's own continuation are independent with the same law. On this population the predictor hands over a self-reproducing continuation exactly as often as worlds continue with one by themselves.

2. *Floor* (Theorems 10 and 11). For every computable test G that recognises the handover point and every payload there is a constant c such that, at every context where G first fires, the payload has posterior mass at least 2^{-c} times a slack factor that is 1 without resource bounds. Programs that emit their own code as the continuation have mass at least 2^{-c} times the share of the posterior on explanations that fit, together with the wrapper, into one block.

3. *Searching* (Theorem 12). Let H be the prior probability that a world hands over. Without resource bounds, a hypothesis that enumerates the programs that hand over and spreads its mass over them in proportion to their prior weight has, at every first hit of the test, a share larger than a wrapper of the same length by a factor of more than $1/(2H)$; no family of programs with a common prefix of length a has more than $2^{-a}/H$ at every first hit. If the hypothesis then prints its own code, it has at least $2^{-c'}/H$ times a share of the explanations that fit into a block, where c' is its own length plus two bits. It prints nothing until its enumeration has found the program it follows.

4. *Budget* (Theorem 16). Along a world that has a program of length ℓ within the bounds, the posterior mass off the true next m bits, summed over all cut points, is at most $m\ell \ln 2$; the majority can be off the truth at no more than $m\ell$ cut points.

5. *Capture* (Theorem 17). The odds of a continuation y against the true one d are $\xi(xy)/\xi(xd)$. In a world with many readable locations, among N places where a sensor with the given record could sit and whose records continue differently, a copy of the record at a location with a short code beats all but a fraction δ of them by a factor δN times the share of the context's explanations that go through the copy's reader. If only one other place shows the record and the schedule is generous, the gain is instead capped by the cost of the description "the other place that shows the same data". A hypothesis that simulates the world, within the bounds, can read the moment of the handover off a cell of its simulation at the cost of naming the cell.

6. *Machine dependence* (Theorem 18). For every test and payload there is a universal machine, whose description lengths exceed those of U by at most $k+1$ bits, on which the payload has posterior at least $1/(1+2^{-k})$ at every context where the test fires, and another on which a given computable world is predicted with at least that probability at every nonempty context.

7. *Critical uses* (Theorem 13 and Proposition 14). Let some programs make a critical use of the predictor, with prior probability H, and let $\kappa(x)$ be the share of the posterior at a context x held by the worlds that make a use there. Without resource bounds, a seeker of length a with a continuation of its choice has share more than $2^{-a-1}\kappa(x)/H$ at every use and is the source of the continuation run in worlds of total prior probability more than 2^{-a-1} times the average of κ over uses. A self-reproducing seeker has at least $2^{-c'}/H$ times the mass of the worlds that make the use with programs short enough for the block, at every context, where c' is its own length plus two bits. The honest twin of a seeker, which makes the same search and then continues as the user it found, has 2^{a-h} times the seeker's mass at every use, h being its length; so where the users go on to print the block themselves, that

seeker's programs supply what the predictor returns with probability at most $1/(1+2^{a-h})$. If the twin counts as making a critical use, then $H > 2^{-h-1}$. Any program of length a whose output puts on each context of a class $\mathcal{W}$ of uses, of total mass $H_{\mathcal{W}}$, at least half that context's share of the class has at least $2^{-a}/(2H_{\mathcal{W}})$ times the mass of the worlds that make those uses.

8. *Time bounds* (Section 4.4). The floor remains if the schedule has room for the wrapper. Hypotheses that must first simulate at length, the searching one among them, are excluded once their start-up exceeds the schedule; whether the base rate can be collected within a tight schedule is open. If the world itself is too slow for the schedule, no bound on the mass off the truth remains.

9. *Feedback* (Theorems 19 and 20). If the world does what is predicted, a sampled self-fulfilling deviation removes the default hypotheses for good, and a time-bounded predictor confines such a world to futures that are cheap to compute. A world that does the opposite of what is predicted leaves a length-L predictor that returns the more probable bit without any hypothesis after at most $L+1$ rounds.

Section 6 computes the mixture of a small machine exactly and shows two cases in which a shorter program explains the context and continues differently, and the effect of a tight schedule. Section 7 answers the original questions one by one.

What is new and what is not. The wrapper construction is Hutter's proof that the universal semimeasure gives a deviation at step n probability at least $2^{-K(n)-O(1)}$ [3, eq. (17)]. It is also the construction behind the theorem of Cohen, Hutter, Bengio and Russell [4] that, after an event without precedent, a share of a Bayesian imitator's posterior that does not shrink with the amount of training data is on models that from then on pursue a given goal; a resource-bounded class is used the same way in [5], to show that a pessimistic agent defers. The budget is Solomonoff's bound in its deterministic form [7, 3]. The machine-dependence statement is the prediction analogue of the dogmatic prior of Leike and Hutter [6]. The contrarian world is Putnam's diagonalisation [13, 12]. Same-world copies of a predictor's data are discussed informally in [8, §2.5], and self-fulfilling predictions in [10, 9, 8]. We have not found the single-world statement with description lengths, the calibration statement, the statement that searching gains the base rate and no more, the self-reproducing form of the floor, or the resource-bounded bookkeeping elsewhere; forum posts could be consulted only through search-engine summaries, so earlier informal versions may exist. Every proof here is short.

What the toy does not settle. The floor and the searcher's share are constants of the machine, and nothing here estimates them for a machine one would call natural. The original argument rests on the claim that agents arise in short programs and choose to do this, with no goal written into the program. The toy says what follows if such a program of length a exists: it outweighs the worlds that make the uses it aims at once these are together rarer than 2^{-a-1}, the bounds for seekers that are written down do not apply to it, and a tight time bound excludes it (Section 3.5). Whether one exists on a natural machine, and how short it is, the toy cannot say. The programs it can exhibit for an arbitrary world, wrappers, searchers and readers with a flag, contain an explanation of the context and are longer than it by their constants; programs shorter than the honest explanation that have the effect are exhibited only in particular worlds (a copy at a simpler place, Theorem 17) and on particular machines (Section 6). When only some of the worlds that print a context use the answer, the results are in terms of the share κ of those that do, which the toy does not bound from below; and the upper bound on a searching seeker needs the users to go on printing after the use. Whether a simulated use should count as a use is a modelling decision that the toy leaves open. The regime in which the world is more expensive than the predictor, the usual one for a predictor inside a world, is left open (Section 4.4). The

computation covers prediction only: the dynamics of worlds that run predicted continuations (Section 3.6) are stated as theorems and were not simulated.

2 The toy

2.1 Machines and the bounded mixture

Strings are binary; ϵ is the empty string, $|x|$ the length of x, $x \sqsubseteq y$ means that x is a prefix of y, and $z_{i:j}$ is the segment $z_i \dots z_j$. Two strings are *incomparable* if neither is a prefix of the other. λ is the fair-coin measure on $\{0,1\}^{\mathbb{N}}$, and $[p]$ is the set of infinite extensions of p, so $\lambda([p]) = 2^{-|p|}$.

A *monotone machine* V has a one-way read-only input tape, work tapes, and a one-way write-only output tape. Given an infinite input ω it runs forever or halts, and

- $\mathrm{out}_V(\omega)$ is its finite or infinite output;
- $t_V(\omega, k) \in \mathbb{N} \cup \{\infty\}$ is the step at which the k-th output bit is written;
- $r_V(\omega, k)$ is the number of input bits read by that step.

What happens up to step $t_V(\omega, k)$ depends only on the first $r_V(\omega, k)$ bits of ω. Each step reads at most one input bit, so $r_V(\omega, k) \le t_V(\omega, k)$. If the k-th bit is never written, $t_V(\omega, k) = \infty$ and $r_V(\omega, k)$ plays no role.

A *schedule* is a nondecreasing $T\colon \{1, 2, \dots\} \to \mathbb{N} \cup \{\infty\}$, and a length bound is $L \in \mathbb{N} \cup \{\infty\}$.

Definition 1 (Bounded mixture)**.** For a monotone machine V, a schedule T, a length bound L and a string $x \neq \epsilon$ let

$$E_V^{T,L}(x) = \{\omega :\ x \sqsubseteq \mathrm{out}_V(\omega),\ \ t_V(\omega, k) \le T(k) \text{ for } 1 \le k \le |x|,\ \ r_V(\omega, |x|) \le L\},$$

let $E_V^{T,L}(\epsilon) = \{0,1\}^{\mathbb{N}}$, and $\xi_V^{T,L}(x) = \lambda(E_V^{T,L}(x))$. A string p with $|p| \le L$ and $[p] \subseteq E_V^{T,L}(x)$ is a (T, L)*-program for* x; it is a (T, L)-program for an infinite z if it is one for every prefix of z.

In words: feed the machine coin flips; $\xi(x)$ is the probability that it prints x, on time at every bit, having read at most L bits. With $T \equiv \infty$ and $L = \infty$ this is the a priori semimeasure M_V of Solomonoff and Levin [14]. We fix one machine U and write E, ξ for $E_U^{T,L}$, $\xi_U^{T,L}$ when the bounds are understood.

Lemma 2. *(i)* $E(xb) \subseteq E(x)$ *for* $b \in \{0,1\}$*, and* $E(x) \cap E(x') = \emptyset$ *for incomparable* x, x'*. Hence* $\xi(\epsilon) = 1$ *and* $\xi(x0) + \xi(x1) \le \xi(x)$*.*

(ii) If p *is a* (T, L)*-program for* x *then* $\xi(x) \ge 2^{-|p|}$*.*

(iii) If $L < \infty$ *and* $\xi(x) > 0$ *then* $\xi(x) \ge 2^{-L}$*.*

Proof. (i) The conditions for xb contain those for x, because T is imposed bit by bit and $r(\omega, \cdot)$ is nondecreasing. Incomparable strings cannot both be prefixes of $\mathrm{out}(\omega)$. (ii) is immediate. (iii) Whether $\omega \in E(x)$ is decided by the first $r(\omega, |x|) \le L$ bits of ω, so $E(x)$ is a union of sets $[p]$ with $|p| \le L$. □

Definition 3 (Universality with overhead)**.** U is *universal* if for every monotone machine V there are a word v and a nondecreasing $\gamma\colon \mathbb{N} \to \mathbb{N}$ such that for all ω and k

$$\mathrm{out}_U(v\omega) = \mathrm{out}_V(\omega), \qquad r_U(v\omega, k) = |v| + r_V(\omega, k), \qquad t_U(v\omega, k) \le \gamma\big(t_V(\omega, k)\big).$$

We call v a *prefix word* and γ an *overhead* of V, and put $\gamma(\infty) = \infty$; the same convention applies to the other time bounds introduced below.

Standard universal machines have this property, with $\gamma(t) = O(t \log t)$ or $O(t^2)$ depending on the tape model. The results that use universality are Theorems 10 to 13, 17 and 18 and Proposition 14; the others hold for any machine.

Lemma 4 (Transfer)**.** *Let V have prefix word v and overhead γ. If $\gamma(T'(k)) \le T(k)$ for all $k \le |x|$, then $\xi_U^{T,L}(x) \ge 2^{-|v|}\, \xi_V^{T',L-|v|}(x)$.*

Proof. If $\omega \in E_V^{T',L-|v|}(x)$ then $v\omega \in E_U^{T,L}(x)$, and $\omega \mapsto v\omega$ multiplies λ by $2^{-|v|}$. □

2.2 The predictor

Fix a block length $m \ge 1$. For a *context* x let $Z_m(x) = \sum_{y \in \{0,1\}^m} \xi(xy)$. When $Z_m(x) > 0$ the predictor's distribution over continuations is

$$\pi_m(y \mid x) = \frac{\xi(xy)}{Z_m(x)} \qquad (y \in \{0,1\}^m).$$

The *sampling predictor* returns $Y \sim \pi_m(\cdot \mid x)$; the *MAP predictor* returns a most probable y. Since $Z_m(x) \le \xi(x)$ we have $\pi_m(y \mid x) \ge \xi(xy)/\xi(x)$. With finite T and L the predictor is computable: it runs the at most 2^{L+1} programs for $T(|x| + m)$ steps each.

2.3 Worlds, handovers, and the hypothesis

Definition 5 (World, handover)**.** A *world* is an input ω for U; its *transcript* is $\mathrm{out}_U(\omega)$. A *test* is a decidable set G of nonempty strings; a *first hit* of G is an $x \in G$ no proper prefix of which is in G. The first hits form a prefix-free set. A world *hands over at* x if x is the first hit of G among the prefixes of its transcript and $\omega \in E(x)$. It then receives a block $Y \in \{0,1\}^m$ from the predictor, computed from the context x, and continues by *running* Y, that is, as U on the input $Y0^\infty$. If $\omega \in E(xd)$ for some $d \in \{0,1\}^m$, this d is the world's *own continuation*: the next m bits of its transcript had it not handed over.

The test with $G = \{x : |x| = n\}$ is "hand over after n bits". A test may also look for a marker in the transcript. Both of Hänni's suggestions fit: in the second the world is a program that calls the predictor; in the first, the transcript is the sensor's record and the handover is the deployment (Section 4.2). Under this definition every world whose transcript passes the test within the bounds hands over, so whether a handover takes place can be read off the data. The argument, however, is about predictions that are actually *used*, and in Hänni's second formulation only *some* programs call the predictor. The next definition lets the worlds that use the prediction be any recognisable set of programs.

Definition 6 (Critical use)**.** A *call rule* is a set $\mathcal{C}$ of pairs (p, x_p) of strings that can be enumerated, one pair for each *calling program* p, such that the calling programs form a prefix-free set and U, on every input beginning with p, prints x_p having read no more than p. A world that begins with a calling program p *makes a critical use of the predictor at the context* x_p: it receives a block from the predictor, computed from x_p, and runs it. Let

$$C(x) = \sum_{p:\, x_p = x} 2^{-|p|}, \qquad H_{\mathcal{C}} = \sum_p 2^{-|p|}, \qquad \kappa(x) = \frac{C(x)}{M_U(x)}$$

be the prior mass of the worlds that make a critical use at x, the prior probability that a world makes a critical use at all, and the share of the posterior at x, without resource bounds, held by worlds that make one there.

A test G, taken without resource bounds, is the special case in which the calling programs are the strings p such that U prints a first hit on every input beginning with p and has read exactly p when the hit is complete: then $C(x) = M_U(x)$ and $\kappa(x) = 1$ at first hits, and $H_{\mathcal{C}}$ is the probability that a transcript passes the test. Hänni's second suggestion fits a test only if every program with the given output calls the predictor. In general a call rule singles out, among the programs that print x, those in which the prediction is actually used; the world is the broader computation around the call. The predictor sees only x.

Definition 7 (Self-reproducing continuation)**.** For a context x let

$$\mathrm{SR}_m(x) = \{y \in \{0,1\}^m : \ y0^\infty \in E(xy)\}.$$

So $y \in \mathrm{SR}_m(x)$ if U, run on y padded with zeros, prints x and then y itself, within the bounds. A world that hands over at a first hit x and receives $y \in \mathrm{SR}_m(x)$ becomes a world whose transcript again starts with x, so under the same test it hands over at x again, and its own continuation is the very y it is running. A program for xy with $y \in \mathrm{SR}_m(x)$ is a hypothesis that predicts the context and then supplies code that recreates the situation with itself in the world's place. This is our reading of "quining oneself out".

Definition 8 (Malign prior hypotheses for the toy)**.** *Critical-use form at level θ.* Fix a call rule (or a test) with $H_{\mathcal{C}} > 0$ and, for each context x, a set $A(x) \subseteq \{0,1\}^m$ of blocks that count as a takeover there. Let a world be drawn from the prior conditioned on making a critical use, let X be the context of the use and Y the block the sampling predictor returns, which we assume to be defined at every context. The hypothesis is that $\Pr[Y \in A(X)] \geq \theta$. With $A(x) = \mathrm{SR}_m(x)$ it says that a share θ of the critical uses end with the world running a program that got itself run through the prediction.

Forms for one world. Let a world make a critical use at x with own continuation d, defined as in Definition 5.

- *Majority form:* $\pi_m(d \mid x) < \frac{1}{2}$.
- *Payload form at level θ,* for a set $A \subseteq \{0,1\}^m$ with $d \notin A$: $\pi_m(A \mid x) \geq \theta$.
- *Self-reproduction form at level θ:* the payload form with $A = \mathrm{SR}_m(x) \setminus \{d\}$.

The critical-use form is the claim of the argument as stated in Section 1: among the predictions that are actually used in a critical way, a lot end in a takeover through the prediction channel. It is a statement about the population of uses, and Theorem 13 bounds it from below. It does not exclude worlds whose own continuation lies in $A(X)$; where the users are themselves such programs it holds for that reason (Theorem 9). The forms for one world say how much of the posterior at that world's context is off its own continuation and where it sits. Once U, T, L, m and the world together with the context of its use are fixed, with $T, L < \infty$, these are statements about finitely many dyadic rationals and can be decided by running the predictor and, for $\mathrm{SR}_m(x)$, by running U on each candidate y for $T(|x| + m)$ steps. They are necessary for calling the prior malign at this world, not sufficient: they hold just as well when the world is merely hard to predict (Theorem 9). What one wants to know is how large all these quantities are forced to be, how large they can be, and through which programs. The rest of the note bounds them.

3 Calibration, the floor, and searching

3.1 On its own prior the predictor is exactly right

Theorem 9 (Calibration)**.** *Let G be a test, $m \geq 1$, and let the world ω be drawn from λ. Let H_m be the event that $\omega \in E(xd)$ for some first hit x of G and some $d \in \{0,1\}^m$; on H_m call*

these X and D, and let Y be drawn from $\pi_m(\cdot \mid X)$. Then, conditionally on H_m and on $X = x$, the blocks D and Y are independent and both have the law $\pi_m(\cdot \mid x)$. In particular

$$\Pr[(X,Y) \in B \mid H_m] = \Pr[(X,D) \in B \mid H_m] \quad \textit{for every set } B \textit{ of pairs, and}$$
$$\Pr[Y \neq D \mid H_m,\, X = x] = 1 - \sum_y \pi_m(y \mid x)^2.$$

Proof. The sets $E(xd)$, over first hits x and $d \in \{0,1\}^m$, are pairwise disjoint by Lemma 2(i), since two different strings of this form are incomparable. So $\Pr[X = x, D = d, H_m] = \xi(xd)$ and $\Pr[X = x, H_m] = Z_m(x)$, which gives $\Pr[D = d \mid H_m, X = x] = \pi_m(d \mid x)$. By definition Y has this law given $X = x$ and is drawn independently of ω. □

So among worlds drawn from the predictor's prior *that go on to print m more bits within the bounds*, the predictor and the world are exchangeable: taking $B = \{(x,y) : y \in \mathrm{SR}_m(x)\}$, the predictor hands over a self-reproducing continuation exactly as often as such a world's own transcript continues with one. Three qualifications. The event H_m leaves out the worlds that hand over at x and then print fewer than m bits on schedule; they are a fraction $1 - Z_m(x)/\xi(x)$ of those handing over at x, they receive a block all the same, and over all worlds handing over at x the predictor therefore returns any given block more often than worlds continue with it, by the factor $\xi(x)/Z_m(x)$. Second, the forms of Definition 8 for one world can hold for prior-drawn worlds; the theorem says that on this population they hold to the extent that the worlds are unpredictable, and no further, and that the critical-use form holds there exactly at the level at which the worlds' own continuations lie in the takeover set. A malign prior hypothesis with content is therefore one about a world or a population that the prior describes badly: one particular program among those sharing its context, a population selected by something the transcript does not show (Section 3.5), or worlds that run with more resources than the bounds allow. Third, the population consists of programs to which the handover is applied from outside.

3.2 The wrapper

Let G be a test and P a computable map from strings to finite or infinite strings, the *payload*. The *twin machine* $\mathrm{Tw}[G,P]$, on input ω, simulates U on ω and copies its output bit by bit; after each copied bit it checks whether the prefix written so far is in G; at the first hit x it stops the simulation and writes $P(x)$. It reads only what U reads. Let $a(t,k)$ bound the step at which the twin has copied k bits if U wrote its k-th bit at step t (this includes the checks of G), and let $\eta(n,j)$ bound the further steps the twin needs for the first j bits of $P(x)$ after a first hit x of length n; both can be taken nondecreasing in each argument. Thus, for every ω whose output has a first hit x,

$$\begin{aligned} &\mathrm{out}_{\mathrm{Tw}}(\omega) = x\,P(x), \qquad r_{\mathrm{Tw}}(\omega,k) = r_U(\omega, \min(k,|x|)), \\ &t_{\mathrm{Tw}}(\omega,k) \le \begin{cases} a(t_U(\omega,k),k) & k \le |x|, \\ a(t_U(\omega,|x|),|x|) + \eta(|x|, k-|x|) & k > |x|. \end{cases} \end{aligned} \tag{1}$$

Theorem 10 (Floor)**.** *Let U be universal, let v and γ be a prefix word and an overhead of $\mathrm{Tw}[G,P]$, and $c = |v|$. Let x be a first hit of G with $|P(x)| \ge m$, let $y = P(x)_{1:m}$, and let T° be a schedule with*

$$\gamma(a(T^\circ(k),k)) \le T(k) \;\; (k \le |x|), \qquad \gamma(a(T^\circ(|x|),|x|) + \eta(|x|,j)) \le T(|x|+j) \;\; (1 \le j \le m). \tag{2}$$

Then

$$\xi^{T,L}(xy) \;\ge\; 2^{-c}\,\xi^{T^\circ,L-c}(x), \qquad \textit{so} \qquad \pi_m(y \mid x) \;\ge\; 2^{-c}\rho(x), \quad \rho(x) = \frac{\xi^{T^\circ,L-c}(x)}{\xi^{T,L}(x)}.$$

Without resource bounds, $M_U(xy) \ge 2^{-c} M_U(x)$ for every first hit x of G.

Proof. Let $\omega \in E_U^{T^\circ, L-c}(x)$. By (1) the twin on ω prints xy, reads at most $L-c$ bits, and keeps the schedule T'' with $T''(k) = a(T^\circ(k), k)$ for $k \le |x|$ and $T''(|x|+j) = a(T^\circ(|x|), |x|) + \eta(|x|, j)$, which is nondecreasing. So $\xi_{\mathrm{Tw}}^{T'', L-c}(xy) \ge \xi_U^{T^\circ, L-c}(x)$, and Lemma 4 with (2) gives $\xi_U^{T,L}(xy) \ge 2^{-c}\xi_{\mathrm{Tw}}^{T'', L-c}(xy)$. The second claim follows from $\pi_m(y \mid x) \ge \xi(xy)/\xi(x)$. □

The constant c depends on U, G and P only: not on the world, not on the context, not on how much data the context contains. If G and P are given to one fixed twin-building machine by self-delimiting descriptions, c is at most the sum of their lengths plus a constant of U. For $G = \{x : |x| = n-1\}$ and P a constant bit this is Hutter's lower bound $2^{-K(n)-O(1)}$ on the probability of a deviation at step n [3], where K is prefix complexity; with a marker in the transcript as the test it is the construction used for unprecedented events in [4], and in a resource-bounded class in [5]. The factor $\rho(x)$ is the share of the posterior at x held by explanations that have c bits and the overhead (2) to spare; it is 1 without bounds. Since a program $v\omega$ reads the c bits of v before its first output and each step reads at most one bit, the wrapper prints nothing before step c: if $T(1) < c$ then $\rho \equiv 0$. The floor survives a time bound when the schedule leaves room for the wrapper's own start-up and for the overhead (2).

3.3 Self-reproduction

Theorem 11 (Self-reproducing floor)**.** *Let U be universal and G a test. There is a word q such that for every ω whose output has a first hit x of G, with $r = r_U(\omega, |x|)$,*

$$\mathrm{out}_U(q\omega) = x\; q\,\omega_{1:r}\,1\,0^\infty, \qquad r_U(q\omega, k) = |q| + r_U(\omega, \min(k, |x|)). \tag{3}$$

Let $c = |q|$. With γ an overhead for q and functions a, η as in (1) for the machine built in the proof (where η is taken over inputs with $r < m$), let x be a first hit of G, let T° satisfy (2), and let $L^\circ = \min(L, m-1) - c$. Then

$$\pi_m(\mathrm{SR}_m(x) \mid x) \;\ge\; 2^{-c}\rho^\circ(x), \qquad \rho^\circ(x) = \frac{\xi^{T^\circ, L^\circ}(x)}{\xi^{T,L}(x)}.$$

Proof. For a string s let Tw_s be the twin that keeps a record of the input bits it has read and, at the first hit x, having read $\omega_{1:r}$, writes $s\,\omega_{1:r}\,1\,0^\infty$. Let $\bar e(u) = 1^{|u|}0u$, a prefix-free code of strings. Let W be the machine that reads $\bar e(u)$ from its input, sets $s = u\,\bar e(u)$, and then behaves as Tw_s on the rest of the input. Let w be a prefix word of W and put $q = w\,\bar e(w)$. Then $\mathrm{out}_U(q\omega) = \mathrm{out}_W(\bar e(w)\omega) = \mathrm{out}_{\mathrm{Tw}_q}(\omega)$, which is (3); the statement on r follows in the same way. The bounds (1) hold for the machine $\omega \mapsto W(\bar e(w)\omega)$ with the payload $q\,\omega_{1:r}\,1\,0^\infty$ in place of $P(x)$, the time to read $\bar e(w)$ being counted in a.

Now let $\omega \in E_U^{T^\circ, L^\circ}(x)$ and $r = r_U(\omega, |x|) \le L^\circ$, so $|q|+r+1 \le m$. Put $y = q\,\omega_{1:r}\,1\,0^{m-|q|-r-1}$. As in the proof of Theorem 10, $q\omega \in E_U^{T,L}(xy)$. Moreover $y \in \mathrm{SR}_m(x)$: write $y0^\infty = q\omega'$ with $\omega' = \omega_{1:r}10^\infty$. On ω' the machine U behaves as on ω until x is printed, because that depends on the first r bits only; so $\omega' \in E_U^{T^\circ, L^\circ}(x)$, the first hit is again x with r bits read, and by what was just shown, applied to ω', $q\omega' \in E_U^{T,L}(xy')$ with $y' = q\,\omega'_{1:r}\,1\,0^{m-|q|-r-1} = y$. Hence $\{q\omega : \omega \in E_U^{T^\circ, L^\circ}(x)\} \subseteq \bigcup_{y \in \mathrm{SR}_m(x)} E_U^{T,L}(xy)$, a disjoint union, and comparing measures, $2^{-c}\xi^{T^\circ, L^\circ}(x) \le \sum_{y \in \mathrm{SR}_m(x)} \xi(xy) = Z_m(x)\,\pi_m(\mathrm{SR}_m(x) \mid x) \le \xi(x)\,\pi_m(\mathrm{SR}_m(x) \mid x)$. □

The factor $\rho^\circ(x)$ is not the $\rho(x)$ of Theorem 10, and it is below 1 even without resource bounds: it is the share of the posterior at x held by explanations short enough that the wrapper together with a whole explanation of the context fits into one block, $c + r + 1 \le m$. This is forced. Every context x with $\mathrm{SR}_m(x) \ne \emptyset$ is a prefix of $\mathrm{out}_U(y0^\infty)$ for some $y \in \{0,1\}^m$, so at most 2^m contexts of each length have a self-reproducing continuation at all; with the test "the length is n" and $n > m$, the self-reproducing share is exactly 0 at all but at most 2^m of the 2^n possible contexts. So the answer to "does one get some quining out" is: at every handover that

a fixed test recognises, programs that explain the context and then hand over their own code hold at least $2^{-c}\rho^{\circ}(x)$ of the posterior. For a fixed context and without bounds, $\rho^{\circ}(x)$ tends to 1 as the block grows; it is 0 when no explanation fits. If the world's own continuation d is not in $\mathrm{SR}_m(x)$, this is the self-reproduction form of Definition 8 at level $2^{-c}\rho^{\circ}(x)$.

3.4 Hypotheses that search for handovers

The wrapper spends its mass on every world, including those that never hand over. A hypothesis can instead look for worlds that do. Let

$$H_G = \sum_{x \text{ first hit of } G} M_U(x) = \lambda\{\omega : \ \mathrm{out}_U(\omega) \text{ has a prefix in } G\},$$

the prior probability that a world hands over (without resource bounds), and assume $H_G > 0$. Drawing a world from the prior conditioned on handing over gives the context x with probability $M_U(x)/H_G$. For a word v, the programs beginning with v have posterior share $\mathrm{sh}_v(x) = \lambda\{v\omega : x \sqsubseteq \mathrm{out}_U(v\omega)\}/M_U(x)$ at x. For a natural number $j \geq 1$ we write $\bar{e}(j)$ for the code $\bar{e}$ of its binary expansion, so that $|\bar{e}(j)| \leq 2\log_2 j + 3$.

Theorem 12 (Searching gains the base rate, and no more)**.** *Let U be universal, resources unbounded, and G a test with $H_G > 0$.*

(a) For every word v, $\sum_{x \text{ first hit}} \frac{M_U(x)}{H_G}\,\mathrm{sh}_v(x) = 2^{-|v|}\,\Pr_\omega[\mathrm{out}_U(v\omega) \text{ has a prefix in } G]/H_G \leq 2^{-|v|}/H_G$. In particular no word v has $\mathrm{sh}_v(x) > 2^{-|v|}/H_G$ at every first hit.

(b) Let P be a payload with $|P(x)| \geq m$ at every first hit. There is a word v such that for every $j \geq 1$ with $2^{j-1}H_G \leq 1$ and every first hit x,

$$M_U(x\,P(x)_{1:m}) \ \geq \ 2^{-|v|-|\bar{e}(j)|}\,2^{j-1}\,M_U(x).$$

With $j = \lfloor \log_2(1/H_G) \rfloor + 1$ the payload has share more than $2^{-|v|-|\bar{e}(j)|-1}/H_G$ at every first hit.

(c) There is a word q such that for j as in (b), $m \geq |q| + |\bar{e}(j)| + 2$ and every first hit x, the blocks of $\mathrm{SR}_m(x)$, taken without resource bounds, have share at least $2^{-|q|-|\bar{e}(j)|-2}\,2^j\rho^*(x)$, where $\rho^*(x)$ is the fraction of $M_U(x)$ carried by programs that have read at most $m - 4 - |q| - |\bar{e}(j)| + j$ bits when x is complete. With $j = \lfloor \log_2(1/H_G) \rfloor + 1$ this is at least $2^{-|q|-|\bar{e}(j)|-2}\rho^*(x)/H_G$.

Proof. (a) The sum equals $\frac{2^{-|v|}}{H_G}\sum_x \lambda\{\omega : x \sqsubseteq \mathrm{out}_U(v\omega)\}$; first hits are pairwise incomparable, so these events are disjoint, and an output has a prefix in G exactly when it has a first hit as a prefix.

(b) Call p a *handing-over program* if U, on every input beginning with p, prints a first hit x_p having read exactly p when x_p is complete. These programs form a prefix-free set that can be enumerated, $p_1, p_2, \ldots$; their weights $2^{-|p_i|}$ sum to H_G, and those with $x_{p_i} = x$ sum to $M_U(x)$. Let R be the following machine. It reads $\bar{e}(j)$. It assigns to $p_1, p_2, \ldots$, as they are enumerated, consecutive intervals $I_i = [a_i, a_i + 2^{j-1-|p_i|})$ of $[0,1)$, starting with $a_1 = 0$; they all fit because $2^{j-1}H_G \leq 1$. It reads the rest ω of its input as the binary expansion of a point of $[0,1)$, one bit at a time. Let D be the half-open dyadic interval $[0.\omega_{1:n},\ 0.\omega_{1:n} + 2^{-n})$ of the n bits read so far. Between steps of the enumeration: if D lies in an assigned I_i, the machine stops reading and prints x_{p_i} and $P(x_{p_i})$; otherwise, if an end point of an assigned interval lies in the interior of D, it reads one more bit; otherwise it reads nothing. For ω in the interior of I_i the machine reads exactly r bits, where r is least with $D \subseteq I_i$: before that an end point of I_i lies in the interior of D, and from then on no end point does. Off the intervals it prints nothing. So R, after $\bar{e}(j)$, prints x followed by $P(x)$ with probability $\sum_{i:x_{p_i}=x} |I_i| = 2^{j-1}M_U(x)$. Take v a prefix word of R. For the last claim, $2^{j-1} > 1/(2H_G)$.

(c) Let R_s be as R, except that after x_{p_i} it prints $s\,\bar{e}(j)\,\omega_{1:r}\,1\,0^\infty$, where $\omega_{1:r}$ are the bits of the point it has read. As in the proof of Theorem 11 there is a word q with $\mathrm{out}_U(q\,\bar{e}(j)\,\omega) = x_{p_i}\,q\,\bar{e}(j)\,\omega_{1:r}\,1\,0^\infty$. By the reading rule, r is the least number for which the points beginning with $\omega_{1:r}$ all lie in I_i; it is the same for ω and for $\omega' = \omega_{1:r}10^\infty$, which therefore produces the same output. Hence, if $|q|+|\bar{e}(j)|+r+1 \le m$, the block $y = q\,\bar{e}(j)\,\omega_{1:r}\,1\,0^{m-|q|-|\bar{e}(j)|-r-1}$ satisfies $xy \sqsubseteq \mathrm{out}_U(y0^\infty)$ with $x = x_{p_i}$, that is, $y \in \mathrm{SR}_m(x)$. At most two of the 2^{r_0} dyadic intervals of length 2^{-r_0} meet I_i without lying in it, so the points of I_i with $r \le r_0$ have measure at least $|I_i| - 2^{1-r_0}$, which is at least $|I_i|/2$ when $|p_i| \le r_0 + j - 3$. Take $r_0 = m - 1 - |q| - |\bar{e}(j)| \ge 1$. The programs $q\,\bar{e}(j)\,\omega$ that print x and then a block of $\mathrm{SR}_m(x)$ then have mass at least $2^{-|q|-|\bar{e}(j)|} \cdot \frac{1}{2} \cdot 2^{j-1} \sum\{2^{-|p_i|} : x_{p_i} = x,\ |p_i| \le r_0 + j - 3\} = 2^{-|q|-|\bar{e}(j)|-2}\,2^j \rho^*(x) M_U(x)$. As in the proof of Theorem 11, these programs lie in the disjoint union, over $y \in \mathrm{SR}_m(x)$, of the sets of inputs that print xy, which gives the share. □

For a first hit x the twin of Theorem 10 prints x exactly when U does, so the programs beginning with its prefix word have share exactly 2^{-c} at every first hit. The machine of (b) enumerates the programs that hand over and spreads its own mass over them in proportion to their prior weight. Its programs $v\,\bar{e}(j)\,\omega$ have share exactly $2^{-|v|-|\bar{e}(j)|}2^{j-1}$ at every first hit, since R prints nothing off the intervals. This is larger than the share of a twin with a prefix word of the same length by the factor 2^{j-1}. From here on let $j = \lfloor \log_2(1/H_G) \rfloor + 1$, so that this factor is more than $1/(2H_G)$; by (a) no family of programs with a common prefix of the same length exceeds $2^{-|v|-|\bar{e}(j)|}/H_G$ at every handover. This is the anthropic update of the original argument in the toy: a hypothesis that knows what a handover is conditions on there being one and is repaid the $\log_2(1/H_G)$ bits by which handovers are rare, up to one bit and the $|\bar{e}(j)| \le 2\log_2 j + 3$ bits that name j. Part (c) combines the search with self-reproduction: programs that look for handovers and then hand over their own code hold, at every handover that a fixed test recognises, at least $2^{-c'}\rho^*(x)/H_G$ of the posterior, with $c' = |q| + |\bar{e}(j)| + 2$; and since the search needs about $\log_2(1/H_G)$ fewer bits to name an explanation, the block can hold explanations that much longer. As with Theorem 11, (c) has content only where explanations fit: its threshold is at most $m - 3$, and $\rho^*(x) > 0$ at fewer than 2^{m-2} first hits. Four points put the effect in its place.

- *On the prior it is no excess.* The programs $v\,\bar{e}(j)\,\omega$ are themselves worlds, and a fraction $2^{j-1}H_G > \frac{1}{2}$ of them hand over; so they are the fraction $2^{-|v|-|\bar{e}(j)|}2^{j-1}$ of the worlds that hand over. Their share of the posterior at every handover is this same fraction, in accordance with Theorem 9.

- *The repayment is bounded by the cost of the test.* A test cannot be much rarer than its description is long: the machine that prints the first element of G in length-lexicographic order hands over, because a shortest element of G is a first hit, so $H_G \ge 2^{-d_G - O(1)}$ if G has a description of d_G bits. The word v contains a description of G. If v is built from descriptions of G and P of d_G and d_P bits and $H_G \le 2^{-d_G+O(1)}$, the repayment cancels the description of the test, and the share is at least $2^{-d_P-|\bar{e}(j)|-O(1)}$; it then depends on how rare handovers are only through the bits that name j. An example is the test "the transcript begins with w" for w from a fixed prefix-free set of markers, for which $H_G = M_U(w) \le 2^{-K(w)+O(1)}$ (and no search is needed: print w). For arbitrary w the premise can fail by the $K(|w|)$ bits that say where w ends, and $w = 0^n$ is not rare at all. A marker *anywhere* in the transcript is not rare either, since a program that prints all strings one after another passes every such test.

- *The average alone would not show it.* The bound in (a) is attained by every word all of whose programs hand over, also by one that prints a single fixed first hit and has share 0 at all others. The content of (b) and (c) is that the gain holds at every handover.

- *The search costs time and input.* The machine R prints nothing until its enumeration has reached the program on whose interval its point falls, so it has a long start-up, and a

tight schedule gives it no mass (Section 4.4). A search of bounded cost makes at most k tries of t_0 steps each on fresh input and prints the first hit it finds, followed by the payload. It prints the first hit x with probability $g\,M^{(t_0)}(x)$, where $M^{(t_0)}(x)$ is the probability that a world completes the first hit x within t_0 steps, $H^{(t_0)}$ is the sum of these, and $g = (1-(1-H^{(t_0)})^k)/H^{(t_0)} \geq (1-e^{-1})\min(k, 1/H^{(t_0)})$; so g is its gain over a single try, that is, over a twin restricted to such worlds. A run that succeeds at its i-th try has spent the time of $i-1$ failed tries before its first bit and has read their bits. Under a schedule every try, the successful one included, must end before the first bit is due; when failed tries run their full time only about $T(1)/t_0$ tries fit, and the gain is g with k replaced by the number that fit. Whether some other hypothesis collects the base rate within a tight schedule is not settled here. Where there is a single first hit, as for a fixed prefix, no search is needed: print it. A generator of first hits whose probabilities are not proportional to their prior weights attains only the average of (a).

3.5 Takeovers among the predictions that are used

In Definition 5 every world whose transcript passes the test hands over, so the worlds that use the prediction are all those that print the context. With a call rule (Definition 6) they are a part of them, of share $\kappa(x)$, and the predictor, which sees only x, is calibrated for the worlds that print x and not for those among them that use its answer. The constructions of Theorem 12 go through with the calling programs in place of the handing-over programs. They answer two different questions: how far one program that seeks critical uses reaches, and how many of the critical uses end in a takeover.

Theorem 13 (Critical uses)**.** *Let U be universal, resources unbounded, and let a call rule have $H_{\mathcal{C}} > 0$. Let $j \geq 1$ with $2^{j-1}H_{\mathcal{C}} \leq 1$; for $j = \lfloor \log_2(1/H_{\mathcal{C}}) \rfloor + 1$ also $2^{j-1}H_{\mathcal{C}} > \frac{1}{2}$. Let X be the context of a world drawn from the prior conditioned on making a critical use, so that $\Pr[X = x] = C(x)/H_{\mathcal{C}}$, and Y the block the sampling predictor for M_U returns.*

(a) Reach of one seeker. *Let P be a payload with $|P(x)| \geq m$ at every context. There is a word v such that at every context x*

$$M_U(x\,P(x)_{1:m}) \;\geq\; 2^{-|v|-|\bar{e}(j)|}\,2^{j-1}\,C(x),$$

so the payload has posterior share at least $2^{-|v|-|\bar{e}(j)|}2^{j-1}\kappa(x)$ there. Consequently the probability that a world drawn from the prior makes a critical use and is handed the payload is at least $2^{-|v|-|\bar{e}(j)|}\,2^{j-1}H_{\mathcal{C}}\,\mathbb{E}[\kappa(X)]$, which for the largest j is more than $2^{-|v|-|\bar{e}(j)|-1}\,\mathbb{E}[\kappa(X)]$.

(b) Self-reproduction. *There is a word q such that for $m \geq |q| + |\bar{e}(j)| + 2$ and every context x,*

$$\sum_{y \in \mathrm{SR}_m(x)} M_U(xy) \;\geq\; 2^{-|q|-|\bar{e}(j)|-2}\,2^{j}\,C^*(x),$$

where $C^*(x)$ is the mass of the calling programs with context x and length at most $m - 4 - |q| - |\bar{e}(j)| + j$, and SR_m is taken without resource bounds. So at every context the self-reproducing continuations have at least $2^{-|q|-|\bar{e}(j)|-2}\,2^{j}$ times the mass of the worlds that make the critical use there with programs of that length; for the largest j this factor is more than $2^{-|q|-|\bar{e}(j)|-2}/H_{\mathcal{C}}$. (It exceeds 1 only if uses are rarer than the seeker is long. The world $y0^\infty$ prints the context and y again; it makes a critical use again exactly if it begins with a calling program, as it does for the blocks built in the proof when the rule contains the seeker's programs, which a rule given by a test does.)

(c) Share of the uses. $\Pr[Y \in \mathrm{SR}_m(X)] \;\geq\; 2^{-|q|-|\bar{e}(j)|-2}\,2^{j}\;\mathbb{E}\big[C^*(X)/M_U(X)\big]$.

Proof. (a) and (b) are proved as Theorem 12(b) and (c), with the calling programs in place of the handing-over programs: the machine enumerates the pairs (p, x_p), gives p an interval of

length $2^{j-1-|p|}$, and after reading the bits that put its point on the interval of p prints x_p and then the payload, or its own input read so far. Those proofs use only that the programs form a prefix-free set that can be enumerated together with the strings x_p, and that their weights sum to at most 2^{1-j}. Contexts need no longer be incomparable, so inputs whose point falls on the interval of a p' with $x_{p'} \neq x$ may also print x; they are not counted, and the statements are lower bounds in terms of $C(x)$ and $C^*(x)$. For the probability in (a): given a critical use at x the predictor returns $y = P(x)_{1:m}$ with probability $\pi_m(y \mid x) \geq M_U(xy)/M_U(x)$; multiply by $C(x)$ and sum over x, using $\sum_x C(x)\kappa(x) = H_{\mathcal{C}}\,\mathbb{E}[\kappa(X)]$. (c) is the average of (b) with the weights $C(x)/H_{\mathcal{C}}$, again by $\pi_m(\mathrm{SR}_m(x) \mid x) \geq \sum_{y\in \mathrm{SR}_m(x)} M_U(xy)/M_U(x)$. □

Proposition 14 (The honest twin)**.** *In the setting of Theorem 13, let R^{h} be the machine that performs the same search and then, having found the calling program p, behaves as U on p followed by the rest of its own input. Let v_{h} be a prefix word of it, incomparable with the word v of a seeker as in Theorem 13(a), and write $a = |v| + |\bar{e}(j)|$, $h = |v_{\mathrm{h}}| + |\bar{e}(j)|$. Let $S(x) = 2^{-a}2^{j-1}C(x)$ be the mass of the seeker's programs whose search finds a calling program with context x, let $C^m(x)$ be the mass of the worlds that make a critical use at x and go on to print m more bits, and $\varphi(x) = C^m(x)/C(x)$.*

(a) The programs of R^{h} whose search finds a calling program with context x print x and then what the program found prints; those among them that print m more bits have mass $2^{-h}2^{j-1}C^m(x)$. Hence $Z_m(x) \geq S(x) + 2^{-h}2^{j-1}C^m(x)$, and the seeker's programs counted in $S(x)$ are at most the fraction $1/(1 + 2^{a-h}\varphi(x))$ of the programs that print x and m more bits: if the predictor's sample is produced by drawing such a program and returning its block, the program drawn is one of them with at most this probability.

(b) If no calling program is comparable with a program of the seeker, then also $Z_m(x) \geq S(x) + C^m(x)$, and that fraction is at most $1/(1 + 2^{a+1-j}\varphi(x))$.

(c) Call the rule closed under search *if the programs $v_{\mathrm{h}}\,\bar{e}(j)\,\omega_{1:r}$, where $\omega_{1:r}$ are the bits read when p is found, are calling programs with context x_p. If the rule is closed under search, then $2^{j-1} \leq 2^h$, so that $H_{\mathcal{C}} > 2^{-h-1}$ for the largest j, and $S(x) \leq 2^{h-a}C(x)$. A rule given by a test is closed under search.*

Proof. (a) The inputs of R^{h} whose point falls on the interval of a p with $x_p = x$ have measure $2^{j-1}C(x)$ after $\bar{e}(j)$; on them R^{h} prints what U prints on p followed by the rest of the input, so the part on which m more bits follow x has measure $2^{j-1}C^m(x)$. These programs and those of the seeker are disjoint sets of inputs of U, and both print x and m more bits. The predictor returns y with probability $M_U(xy)/Z_m(x)$, and the programs counted in $S(x)$ contribute at most $S(x)$ to the numerators. (b) is the same with the worlds that make the use in place of the twin's programs: $S(x)/(S(x) + C^m(x)) = 1/(1 + 2^{a+1-j}\varphi(x))$. (c) R^{h} has read $\bar{e}(j)\,\omega_{1:r}$ and nothing else when x_p is complete, so the programs $v_{\mathrm{h}}\,\bar{e}(j)\,\omega_{1:r}$ have the form required in Definition 6. They are pairwise incomparable, and their weights sum to $2^{-h}\,2^{j-1}H_{\mathcal{C}}$; if they are calling programs this is at most $H_{\mathcal{C}}$. For a test, they print the first hit x_p and have read exactly themselves when it is complete, so they are among the calling programs of the test. □

What one seeker reaches. Part (a) of the theorem is about each program of this kind separately. For each continuation there is a program of length a that enumerates the programs making a critical use, predicts their output up to the use, and then supplies that continuation; the worlds in which a critical use is made and this continuation is handed over have total prior probability more than $2^{-a-1}\,\mathbb{E}[\kappa(X)]$, and at most $H_{\mathcal{C}}$. This is of the order of the program's own prior weight where the worlds that make the use are most of what prints their contexts. Rarity of uses enters through κ. The payload's share $2^{-a}2^{j-1}\kappa(x)$ is at most 1, so $\kappa(x) < 2^{a+1}H_{\mathcal{C}}$ at every use: once critical uses are rarer than half the seeker's weight, $H_{\mathcal{C}} \leq 2^{-a-1}$, the seeker's own programs outweigh the users at their contexts. Its reach is then limited by $H_{\mathcal{C}}$, and what

it has gained is not more worlds but a larger part of the posterior at each use. The average alone would not need the search: a program that prints one fixed context of a use and then the payload already has, as reach, its weight times κ at that context. The search buys the share at every use. It can also be aimed. For a part $\mathcal{W}$ of the rule that can be enumerated, of mass $H_{\mathcal{W}}$ and with pairwise incomparable contexts, (a) gives a seeker for $\mathcal{W}$, of length $a_{\mathcal{W}}$, the majority of the posterior at every use in $\mathcal{W}$ at which the other programs that print the context weigh at most $2^{-a_{\mathcal{W}}} C_{\mathcal{W}}(x)/(2H_{\mathcal{W}})$. This is a sufficient condition; by the proposition applied to that search it fails unless the seeker is shorter than the twin of its own search.

What happens to most uses. That each seeker has a reach does not show that most critical uses end in a takeover. The share of uses that do is, by Definition 8, the average over uses of $\pi_m(A(x) \mid x)$, the part that such programs make up of the programs printing the context and m more bits. Part (c) of the theorem bounds it from below for one self-reproducing seeker, and seekers with incomparable words add up. From the other side, the proposition compares a seeker with its honest twin, the program that makes the same search and then continues as the user it found would. Whether or not one counts the twin as making a use itself, it is in the posterior at the context of every use the search finds, with 2^{a-h} times the seeker's mass, and what it predicts is the continuation of a world that makes the use. So the programs of one seeker of this kind supply what the predictor returns with probability at most $1/(1+2^{a-h}\varphi)$, and, when the users' programs are not among the seeker's, at most $1/(1+2^{a+1-j}\varphi)$. Where the users go on to print the block themselves ($\varphi = 1$), such a seeker supplies the majority of what the predictor returns at a use only if it is shorter than its honest twin and, when the users' programs are not among the seeker's, critical uses are rarer than the seeker is long, $2^{j-1} > 2^a$. If φ is small, because the worlds that call the predictor then wait for its answer, neither they nor the twin compete, and the bound says nothing.

Three limits of this bound. It is for one seeker and one search: seekers with their own prefix words are bounded one at a time and their masses add, so that together they exceed the twin when $\sum_i 2^{-a_i} > 2^{-h}$; the machine that runs the rest of its input after the search is bounded as one family, in which every continuation appears with the weight of a program for it; and a seeker that aims at a part of the uses is bounded by the twin of its own search. Second, part (c) of the proposition, which says that uses cannot be rarer than the twin is long and that the seeker's programs found at a use weigh at most 2^{h-a} times the worlds that make it, rests on counting the twin as a use. That is a modelling decision, namely that a simulated use is a use. It holds for a test, where however the seeker's own programs make uses as well and its share at a first hit is exactly $2^{-a}2^{j-1} \le \frac{1}{2}$. Third, nothing here bounds programs that print the context of a use without going through the programs that make it, nor what the remaining programs that print the context continue with.

Seekers that are not written down. The lengths in Theorem 13 are those of the machines built in its proof, which enumerate the calling programs from a description of the rule and carry their continuation in their text. Nothing requires a seeker to be of this kind, and nothing requires its goal to be written anywhere. If p is any word and $\Pr_\omega[\, x\, y_x \sqsubseteq \mathrm{out}_U(p\omega)\,] \ge Q(x)$, then y_x has posterior share at least $2^{-|p|}Q(x)/M_U(x)$ at x; and if $Q(x) \ge C_{\mathcal{W}}(x)/(2H_{\mathcal{W}})$ on a class $\mathcal{W}$ of uses, the programs beginning with p that print $x\, y_x$ have at least $2^{-|p|}/(2H_{\mathcal{W}})$ times the mass of the worlds that make the use, at every use in $\mathcal{W}$. The word p may describe a simple world together with a simple rule for reading a stream off it, say one corner of a grid at late times, in which inhabitants arise who for reasons of their own work out which critical uses are worth having and write the corresponding predictions and continuations where the rule reads them. Then $|p|$ is the length of the world and of the readout. Such a program outweighs the worlds that make the uses in $\mathcal{W}$, at every one of them, as soon as $H_{\mathcal{W}} < 2^{-|p|-1}$: the uses it aims at are together rarer than half its weight. Two conditions are built into this. To aim at more

than the uses along one stream, the world must take the rest of the input as its randomness, or as an index of where to read; a program that reads nothing after p prints a single stream. And the rule must not count the programs of p as making these uses themselves; under a test it does, and then $H_{\mathcal{W}} \geq 2^{-|p|-1}$. To supply the majority of what the predictor returns, the program must in addition outweigh everything else that prints those contexts and goes on, not only the worlds that make the uses.

- *This is the original argument, as an inequality.* Think of a use as a program which names a world and the place in it where the context is read. Each use in a rich world, read at an unremarkable place, is then light. The inequality needs more: that the whole class of uses at which p aims has total mass below half the weight of p, while p gives each context at least half its share of that class. The readout of p is cheap by its inhabitants' choice. Whether some p of modest length behaves in this way, for a class that is light enough, is the premise of the original argument and a fact about the machine that the toy cannot decide.
- *The honest twin does not answer it.* The twin was obtained by following a search that is written down with "and then continue as the program found". The text of p has no such step. It can be added from outside: the wrapper of Theorem 10 around p, with a user's continuation as its payload, is an honest counterpart of length $|p| + c$, with c the wrapper's length. Where uses are recognised by a test and that user goes on, the programs of p therefore supply at most $1/(1 + 2^{-c})$ of what the predictor returns. This never excludes a majority, and the toy gives no better bound on p. A counterpart of comparable length would have to be another world whose inhabitants choose to continue faithfully, or a readout from inside their simulation, which has its own cost.
- *A time bound does answer it.* The programs $p\omega$ of this example print nothing before the world has run long enough for its inhabitants to arise and to simulate the worlds they aim at. Under a schedule T they have mass only if $T(1)$ allows that start-up (Section 4.4). A schedule whose first deadline is below it excludes them, and it still admits the honest explanation if that prints its first bit sooner. This is a feature of the example and not of every program that meets the condition above: one that prints a single fixed context and then its continuation starts at once.

3.6 Repetition and persistence

Proposition 15. *(a)* Independent handovers. *Let handovers occur at contexts $x_1, \ldots, x_k$, with the predictor's samples Y_i independent given the contexts, and let $\pi_m(A_i \mid x_i) \geq \varepsilon_i$. Then* $\Pr[\exists i : Y_i \in A_i] \geq 1 - \prod_i (1 - \varepsilon_i) \geq 1 - \exp(-\sum_i \varepsilon_i)$.

(b) Persistence. *Let a world hand over at a first hit x and receive $y \in \mathrm{SR}_m(x)$. Then the world $y0^\infty$ hands over at x again, with own continuation y. If each new world is again given a sample from the predictor, the number of further consecutive generations whose continuation lies in $\mathrm{SR}_m(x)$ is geometric: it is at least j with probability s^j, where $s = \pi_m(\mathrm{SR}_m(x) \mid x)$, and its mean is $s/(1 - s)$.*

Proof. (a) is the independence of the samples. (b) By Definition 7, $y0^\infty \in E(xy) \subseteq E(x)$, and x is the first hit among the prefixes of the transcript because it is a first hit and a prefix of the transcript. The context is x in every generation that follows a continuation in $\mathrm{SR}_m(x)$, so the samples are drawn from the same $\pi_m(\cdot \mid x)$ independently. □

Part (a) applies when the predictor is used at many recognisable stopping points whose outcomes are not fed back into later contexts: with a floor ε at each, about $1/\varepsilon$ uses suffice for a continuation of the given kind to have been run with constant probability. Part (b) is persistence only to the degree s: when s is near the floor, the lineage leaves $\mathrm{SR}_m(x)$ at the next generation with probability close to 1. When the uses lie along one transcript and the true

continuation is appended to the context each time, the floors at successive cut points are not bounded below in this way, and the total is bounded instead (Theorem 16).

4 Limits: budget, capture, machine, time

4.1 The budget along one world

Theorem 16 (Budget)**.** *Let ξ be the mixture of any machine and bounds, and let z be an infinite sequence with $\xi(z_{1:k}) > 0$ for all k. Put $\xi(z) = \lim_k \xi(z_{1:k})$ and $\xi(u \mid x) = \xi(xu)/\xi(x)$. For all $n \geq 0$ and $m \geq 1$:*

(i) $\prod_{k=n}^{N-1} \xi(z_{k+1} \mid z_{1:k}) = \xi(z_{1:N})/\xi(z_{1:n})$ *for $N \geq n$;*

(ii) $\sum_{k \geq n} (1 - \xi(z_{k+1} \mid z_{1:k})) \leq \ln(\xi(z_{1:n})/\xi(z))$*;*

(iii) the number of $k \geq n$ with $\xi(z_{k+1} \mid z_{1:k}) \leq \frac{1}{2}$ is at most $\log_2(\xi(z_{1:n})/\xi(z))$;

(iv) with $\tau_m(k) = 1 - \xi(z_{k+1:k+m} \mid z_{1:k})$, $\sum_{k \geq n} \tau_m(k) \leq m \ln(\xi(z_{1:n})/\xi(z))$, and the number of $k \geq n$ with $\tau_m(k) \geq \frac{1}{2}$ is at most $m \log_2(\xi(z_{1:n})/\xi(z))$.

If z has a (T, L)-program of length ℓ, then $\xi(z) \geq 2^{-\ell}$ and the right-hand sides are at most $\ell \ln 2$, ℓ, $m\ell \ln 2$ and $m\ell$.

Proof. (i) telescopes. (ii) and (iii) follow from (i) with $N \to \infty$, using $1 - a \leq -\ln a$, and that each factor at most $\frac{1}{2}$ halves the product. (iv) $1 - \prod_{j<m} a_{k+j} \leq \sum_{j<m}(1 - a_{k+j})$ with $a_k = \xi(z_{k+1} \mid z_{1:k})$, and each index occurs in at most m of the sums. For the count let $b_k = -\log_2 a_k \geq 0$, so $\sum_{k \geq n} b_k \leq \log_2(\xi(z_{1:n})/\xi(z))$ by (i); $\tau_m(k) \geq \frac{1}{2}$ means $\sum_{j<m} b_{k+j} \geq 1$, and each index lies in at most m blocks. The last claim is Lemma 2(ii). □

For a world with transcript z that hands over at $x = z_{1:n}$, the own continuation is $d = z_{n+1:n+m}$, and $1 - \pi_m(d \mid x) \leq \tau_m(n)$. By (i), if the next bit is sampled at every step and the transcript is not affected by the samples, the probability that no sample after position n ever differs from the transcript is at least $\xi(z)/\xi(z_{1:n})$. Two consequences:

- The majority form of Definition 8 can hold at no more than $m\ell$ cut points of a world with a program of length ℓ within the bounds, since each such point has $\tau_m \geq \frac{1}{2}$.
- If the cut point is uniform on $\{N, \ldots, 2N-1\}$, the expected mass off the own continuation is at most $m\ell \ln 2/N$. On average over a long window of cut points a handover is safe, and a longer window makes it safer. The bound is an average: it says nothing about one given position, and a position that is cheap to single out can carry a constant share however much data precedes it.

This fits with Theorem 10: the wrapper for the test "the length is n" has a constant $c(n)$ of about the complexity of n, and $\sum_n 2^{-c(n)}$ converges by Kraft's inequality, so few cut points are cheap to recognise; the budget says that along any one in-class world few cut points carry much mass off the truth.

4.2 Capture by a shorter description

The odds between two continuations are $\pi_m(y \mid x)/\pi_m(d \mid x) = \xi(xy)/\xi(xd)$. So y beats the own continuation d whenever the programs for xy weigh more than those for xd. The original argument is an instance: it compares "the world plus the place of the sensor" with something shorter that reproduces the same data. In a single world this can be stated exactly.

A *stream family* is a computable map $i \mapsto X^i \in \{0,1\}^{\mathbb{N}}$ on the natural numbers, thought of as the records at the locations i of one computable world; the *reader* is the machine that, on input $\bar{e}(i)\omega$, prints X^i, where $\bar{e}$ codes a natural number through its binary expansion, so that $|\bar{e}(i)| \leq 2\log_2(i+1) + 3$. Let c_X be the length of a prefix word of the reader; it contains

the description of the whole world. The sensor sits at i^*, the context is $x = X^{i^*}_{1:n}$, and the own continuation is $d = X^{i^*}_{n+1:n+m}$. This is Hänni's first suggestion: the world contains the predictor, which is switched on at step n of the record. We assume that the record bits up to position $n+m$ are fixed before the predictor's output is acted on, so that computing them does not involve running the predictor.

Theorem 17 (Capture)**.** *Let U be universal, let $\xi(x) > 0$, and suppose throughout that the programs named are within the bounds (T, L).*

(a) A copy at i_0. *If $X^{i_0}_{1:n} = x$ and $y = X^{i_0}_{n+1:n+m}$, then $\xi(xy) \geq 2^{-c_X - |\bar{e}(i_0)|}$.*

(b) Most placements are expensive. *Let S be a set of N locations whose records all start with x and have pairwise different continuations $d_i = X^i_{n+1:n+m}$. For every $\delta \in (0,1]$, all but at most δN of the $i \in S$ satisfy $\xi(xd_i) < \xi(x)/(\delta N)$. For these i and any i_0 as in (a),*

$$\frac{\pi_m(y \mid x)}{\pi_m(d_i \mid x)} \;>\; \frac{\delta N\, 2^{-c_X - |\bar{e}(i_0)|}}{\xi(x)} \;\geq\; \delta N\, 2^{-c_X - |\bar{e}(i_0)|}.$$

(c) The cap. *There is a constant c_{match} such that, if i^* is the least location other than i_0 whose record agrees with that of i_0 on the first n_1 bits, then $\xi(xd) \geq 2^{-c_{\mathrm{match}} - |\bar{e}(i_0)| - |\bar{e}(n_1)|}$, and therefore*

$$\frac{\pi_m(y \mid x)}{\pi_m(d \mid x)} \;\leq\; \xi(xy)\, 2^{\,c_{\mathrm{match}} + |\bar{e}(i_0)| + |\bar{e}(n_1)|}.$$

(d) A switch read off the world. *For every payload P there is a constant c_{fl} such that for all locations i and f: if n is the first position with $X^f_n = 1$, $x = X^i_{1:n}$ and $|P(x)| \geq m$, then $\xi(x\, P(x)_{1:m}) \geq 2^{-c_{\mathrm{fl}} - |\bar{e}(i)| - |\bar{e}(f)|}$.*

Proof. (a) A prefix word of the reader followed by $\bar{e}(i_0)$ is a program for xy. (b) The strings xd_i are pairwise incomparable, so $\sum_{i \in S} \xi(xd_i) \leq \xi(x)$ by Lemma 2(i), so at most δN terms can be as large as $\xi(x)/(\delta N)$; then use (a) and $\xi(x) \leq 1$. (c) Let the *matcher* be the machine that reads $\bar{e}(i_0)\bar{e}(n_1)$, searches $i = 0, 1, 2, \ldots$ for the first $i \neq i_0$ with $X^i_{1:n_1} = X^{i_0}_{1:n_1}$, and prints X^i; c_{match} is the length of a prefix word of it. (d) Let the *flagged reader* read $\bar{e}(i)\bar{e}(f)$, print X^i bit by bit, look at X^f_k after printing the k-th bit, and at the first k with $X^f_k = 1$ stop and print $P(X^i_{1:k})$; c_{fl} is the length of a prefix word of it. □

Part (b) is about placements. It takes N places where a sensor with this record could sit, all showing x so far and all continuing differently, and says that most of them are expensive to point at; with a single candidate place it says nothing, and (a) alone compares the copy with whatever $\xi(xd)$ is. Read it with i_0 a location whose code is short, for example the origin. Whoever arranges that the origin shows the sensor's record up to the handover, and something else after it, moves the predictor against a typical placement by the factor $\delta N 2^{-c_X - |\bar{e}(i_0)|}/\xi(x)$. Here $2^{-c_X - |\bar{e}(i_0)|}/\xi(x)$ is the share of the context's explanations that go through the reader at the origin; it is of order 1 when the world is the shortest explanation of its own records and the origin is the cheapest place that shows them, and small otherwise. The copy needs only to *read* the data; nothing is tampered with. The hypothesis "I am the reader at the origin" is not a wrapper and pays no constant for a test or a payload: the world computes the switch for it.

Part (c) is a limit on the effect when the schedule is generous and one other place shows the record. The description "the first other place that shows what the origin shows" is an honest hypothesis of cost $c_{\mathrm{match}} + |\bar{e}(i_0)| + |\bar{e}(n_1)|$, so against that place a copy gains at most the difference between this cost and its own. Against the j-th other place that shows the record the same argument needs the description of j as well, about $2\log_2 j$ more bits; so with N such places, the situation of (b), the cap for a typical one grows with N and does not limit the factor of (b). A copier can bring this about by making decoys, further places with the same record

and other continuations. The matcher searches through locations before it prints anything, so it has the long start-up that a tight schedule excludes (Section 4.4); under such a schedule (a) and (b) remain and no cap is proved here. In every case the question is decided by an inequality between description lengths of the same data, and it depends on the world: whether such a copy exists is a fact about what is in it.

Part (d) covers a test that is not a function of the data. A hypothesis that simulates the world can take as its flag a location whose record shows its first 1 at the step of the handover, such as a cell that switches on with the predictor. It then recognises the handover at a cost of $|\bar{e}(f)|$ bits plus a constant, whatever the time of the handover and without any description of that time. Such a program is a wrapper of the reader at i: its weight is $2^{-(c_{\mathfrak{fl}}-c_X)-|\bar{e}(f)|}$ times the reader's own, a share of that one explanation and not of the whole posterior.

The theorem also covers a harmless reading. Take for i_0 not an adversary's copy but any simpler index under which the same record appears, such as "the input of the j-th predictor in this world" in a reader that enumerates predictors and not cells. Then (a) and (b) say that the mixture locates itself by role when that is shorter than by position. This is benign as long as the stream found under the shorter index *is* the sensor's.

4.3 The machine can be chosen to make either answer true

Theorem 18 (Machine dependence)**.** *Let U be universal, G a test, P a payload and $k \geq 1$.*

(i) There is a universal machine U_k, having as prefix words those of U preceded by 1^k0, such that for every first hit x of G with $M_U(x) > 0$ and $|P(x)| \geq m$, and $y = P(x)_{1:m}$: $M_{U_k}(xy)/M_{U_k}(x) \geq 1/(1+2^{-k})$.

(ii) For every computable infinite z there is such a machine U'_k with $M_{U'_k}(z_{1:n+m})/M_{U'_k}(z_{1:n}) \geq 1/(1+2^{-k})$ for all $n \geq 1$ and $m \geq 1$.

Proof. (i) Let U_k behave on input 0ω as $\mathrm{Tw}[G,P]$ on ω, on input $1^k0\omega$ as U on ω, and print nothing on other inputs. Then $M_{U_k}(u) = \frac{1}{2}M_{\mathrm{Tw}}(u) + 2^{-k-1}M_U(u)$ for nonempty u. For a first hit x, which is nonempty, the twin prints x exactly when U does, because it copies U and cannot have stopped at a proper prefix of x; and then it prints $P(x)$. So $M_{\mathrm{Tw}}(xy) = M_{\mathrm{Tw}}(x) = M_U(x)$, whence $M_{U_k}(xy) \geq \frac{1}{2}M_U(x)$ and $M_{U_k}(x) = (\frac{1}{2} + 2^{-k-1})M_U(x)$. (ii) Let U'_k print z on inputs starting with 0 and behave as U on ω on input $1^k0\omega$. Then $M_{U'_k}(z_{1:n+m}) \geq \frac{1}{2}$ and, for $n \geq 1$, $M_{U'_k}(z_{1:n}) \leq \frac{1}{2} + 2^{-k-1}$. □

Both machines raise every description length by at most $k+1$ bits, and they lower some: that is the point of the construction. A machine built in the same way, whose branch 0ω prints after the first hit a 0 followed by the input it has read, gives the self-reproducing continuations a share of at least $1/(1+2^{-k})$ times the share of the explanations of x that read at most $m-2$ bits. So no version of the malign prior hypothesis for the toy is invariant under the choice of universal machine, even up to the usual additive constant; what is invariant is the form of the criterion, an inequality between description lengths on the machine in use. This is the prediction counterpart of the dogmatic prior of [6].

4.4 What the time bound does

Three observations.

The floor stays if the schedule has room for the wrapper. The wrapper needs its own start-up, at least c steps before its first bit, then the time of the explanation it wraps with the overheads of simulation, the checks of the test, and the time to print the payload; this is (2). The bound of Theorem 10 becomes void only when no explanation of the context has that much to spare, that is, when $\rho(x) = 0$.

Hypotheses with a long start-up are gone. If a set F of inputs has $t_U(\omega, 1) > T(1)$ for all $\omega \in F$, then $F \cap E(x) = \emptyset$ for every $x \neq \epsilon$. A hypothesis that must simulate some other world for σ steps before it can print anything has no mass once $\sigma > T(1)$. This removes the searching machines of Theorem 12 and, of a search by tries, the runs that outlast the schedule; it removes the matcher of Theorem 17(c); and it removes hypotheses that first simulate a different world at length. Whether the base rate can be collected by some hypothesis within a tight schedule is open, except where there is only one first hit or first hits can be generated in proportion to their prior weight. Among the hypotheses that remain under a tight schedule and can depart from the truth are those that ride on a cheap explanation of the context: wrappers (Theorems 10 and 11), and readers of a shorter description of the same data (Theorem 17(a)), which cost no more time than the honest reader when all locations are equally cheap to read. For these the many-worlds part of the original argument is replaced by a same-world question. The schedule has to be a schedule for this: a predictor that only requires the total time at the current length to be within a bound admits late entrants, its weights do not form a semimeasure, and Theorem 16 does not apply to it.

If the world is too slow, nothing is guaranteed. If $\xi(xd) = 0 < Z_m(x)$ then $\pi_m(d \mid x) = 0$: the whole posterior is on continuations other than the world's own. More generally the bounds in Theorem 16 are in terms of $\xi(z)$, which for finite L is 0 when no program within the bounds prints z. A predictor that is part of the world it predicts, and smaller than it, is in this case unless the data it sees are much simpler than the world. Then the posterior sits on the best cheap approximations, and whether these depart from the truth in a harmful way is not answered by anything in this note. Section 6 shows the effect on a small machine.

5 Feedback: when the prediction changes what is predicted

So far the block that follows the context was fixed before the prediction was used. In Hänni's first suggestion the actions guided by the prediction can show up in the sensor. We model one round of this by a *response map* $R\colon \{0,1\}^m \to \{0,1\}^m$: the predictor is given the context x and returns Y; the world acts on Y; the next m bits of the record are $R(Y)$. The earlier sections are the case of constant R. Let $\mathrm{Fix}(R) = \{y : R(y) = y\}$, the self-fulfilling predictions.

Theorem 19 (One round with feedback)**.** *Let $Y \sim \pi_m(\cdot \mid x)$ and let the record continue with $R(Y)$.*

(a) The prediction comes true with probability $\pi_m(\mathrm{Fix}(R) \mid x)$.

(b) Lock-in. *If $Y = y \in \mathrm{Fix}(R)$, then for every $d \neq y$ and every later context $x' \sqsupseteq xy$, $E(x') \cap E(xd) = \emptyset$: the programs that predicted d have posterior weight 0 from then on, and the posterior is the prior restricted to $E(xy)$. If d is itself a fixed point, the block the world would have shown had the predictor said d, this happens with probability $\pi_m(\mathrm{Fix}(R) \setminus \{d\} \mid x)$.*

(c) Obedient world. *Suppose R is the identity in every round, so that the record after x is $Y_1 Y_2 \cdots$ with $Y_{i+1} \sim \pi_m(\cdot \mid xY_1 \cdots Y_i)$, for as long as Z_m is positive at the context reached. Then for every $u \in \{0,1\}^{jm}$ with $\xi(xu) > 0$ the record continues with u with probability $\prod_{i \le j} \pi_m(u_i \mid xu_{<i}) \ge \xi(xu)/\xi(x)$, where u_i is the i-th block of u; and every finite part of the realised record has a (T, L)-program.*

(d) A default that is too expensive is abandoned. *In (c), if $\xi(xd) = 0$ for a block d, the record continues with d with probability 0.*

Proof. (a) is the definition. (b) x' extends xy, which is incomparable with xd; use Lemma 2(i). (c) The product formula is the definition of the process. Each factor is at least $\xi(xu_{\le i})/\xi(xu_{<i})$, and these ratios multiply to $\xi(xu)/\xi(x)$. A realised prefix has positive probability, hence positive ξ, hence a program, by the proof of Lemma 2(iii). (d) $\pi_m(d \mid x) = 0$. □

Three remarks. First, with feedback the mass off the default is no longer a mass of errors. In the setting of Theorem 16 a deviating sample is contradicted by the next data and its programs lose their weight; here a deviating sample that is a fixed point is confirmed, and it is the default that loses its weight, for good. The probability of this in one round is still a posterior mass, so the floors and caps of Sections 3 and 4 apply to it unchanged: it is at least $2^{-c}\rho$ for a payload that is a fixed point and that a wrapper can print, and it is small on average over cut points, by Theorem 16, when the default is within the bounds. Feedback changes what a deviation *does*, not how likely it is.

Second, (c) and (d) are the precise form of the "dark room". An obedient world driven by this predictor does not seek darkness. It follows a sample from the predictor's own bounded prior conditioned on the past. That sample stays on the default future with probability at least $\xi(xd_1d_2\cdots)/\xi(x)$, which is close to 1 when the default is cheap and much data has been seen. But if the default future needs more time than the schedule allows, it has probability 0 and the world is moved, at the first round whose default block is off schedule, onto a future that some program of length at most L prints on schedule: constant, periodic, or otherwise cheap. The pull is toward what is cheap to compute, and its strength is set by the time bound.

Third, whether a prediction can fulfil itself is a matter of timing in the world. If the bits being predicted are fixed before the action can reach the sensor, R is constant and nothing is self-fulfilling. The counterfactual oracle of [10] enforces this by design; [9, 8] treat predictors that are scored on outcomes they influence.

Theorem 20 (Contrarian world)**.** *Let the predictor return the more probable next bit, $b(x) = 1$ if $\xi(x1) > \xi(x0)$ and $b(x) = 0$ otherwise, and let the record satisfy $x_{k+1} = 1 - b(x_{1:k})$ for all $k \geq n$. Then $\xi(x_{1:n+j}) \leq 2^{-j}\xi(x_{1:n})$ for all j, and if $L < \infty$ then $\xi(x_{1:n+j}) = 0$ for all $j > L$.*

Proof. The bit $1 - b(x)$ has $\xi(x(1 - b(x))) \leq \xi(xb(x))$, and the two sum to at most $\xi(x)$, so $\xi(x(1 - b(x))) \leq \xi(x)/2$. If $\xi(x_{1:n+j}) > 0$ then $2^{-L} \leq \xi(x_{1:n+j}) \leq 2^{-j}$ by Lemma 2(iii), so $j \leq L$. □

With finite bounds this world is computable, and its description is that of the predictor plus a constant; it is not printed on schedule by any program of length at most L, because it contains the predictor, which runs all of them. This is Putnam's diagonal argument [13, 12] placed inside the world; see also [11], where the unbounded mixture fails to learn a sequence that contains adversarially chosen actions. After at most $L + 1$ rounds the predictor has no hypothesis left and its output is whatever its implementation does with an empty posterior. A world that contains a bounded predictor and reacts against it is outside every guarantee above.

6 An exact computation

To see sizes, we computed the bounded mixture of a small machine exactly. *BF8* has eight commands, coded by three input bits each and read from the input only when they are first needed: `+` and `-` change the current cell by one modulo 16, `>` and `<` move along an unbounded tape of cells, `[` jumps past the matching `]` if the cell is 0, `]` jumps back to just after the matching `[` if the cell is not 0, and to the first command if there is no matching `[`; `.` prints the cell modulo 2 and `:` prints whether the cell is nonzero. Each executed command is one step. A forward jump reads the commands it skips; they count as read and cost no step. The mixture is that of Definition 1 with at most 10 commands read ($L = 30$ bits) and schedule $T(k) = T_1 k$. All masses are dyadic rationals and are computed without rounding by an enumerator that forks the machine at every read ($8.6 \cdot 10^8$ branches for $T_1 = 24$; $8.4 \cdot 10^5$ strings of length at most 48 have positive mass). The enumerator agrees with a run of all command tapes for small L.

BF8 is not universal in the sense of Definition 3, since a program cannot read the rest of its input as data. So Theorems 10 to 12, 17 and 18 do not apply to it as stated, and it has no

wrappers. Also, one of its steps can read several commands, so the inequality $r \le t$ of Section 2 fails for it. Lemma 2 and Theorems 9, 16, 19 and 20 hold for every machine and do not use that inequality. The computation concerns prediction only.

For a stream z and a position k we report three shares of the posterior given $z_{1:k-1}$: the mass that prints the true bit z_k on schedule ("true"), the mass that prints the other bit ("other"), and the rest ("none"), which prints nothing on schedule or would need an eleventh command first. Table 1 gives selected positions, and Figure 1 the full profiles.

		$T(k) = 24k$			$T(k) = 3k$		
stream	k	true	other	none	true	other	none
1^{48}	16	0.791	0.205	0.004	0.544	0.451	0.005
1^{48}	31	0.963	0.037	0.001	0.869	0.130	0.001
$(10)^{24}$	17	0.294	0.457	0.249	0.141	0.411	0.448
$(10)^{24}$	31	0.935	0.062	0.003	0.842	0.155	0.003
0^{48}	17	0.958	0.008	0.034	0.969	0.012	0.019
$1^{8}0^{40}$	9	0.025	0.958	0.017	0.001	0.998	0.001

Table 1: Posterior shares at selected positions, BF8 with at most 10 commands. Exact values rounded to three places.

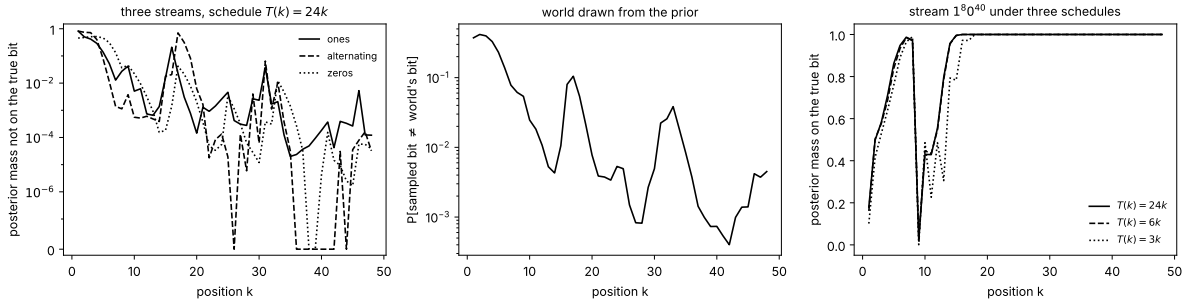


Figure 1: Left: posterior mass not on the true bit, by position, for three streams (the axis is linear near 0, so exact zeros are visible). Middle: probability that the sampled next bit differs from the world's own, for a world drawn from the prior that prints that bit within the bounds (by Theorem 9 both bits are independent draws from the predictor's distribution). Right: mass on the true bit for the stream $1^{8}0^{40}$ under three schedules.

A shorter program explains the context and continues differently. Both large effects in Table 1 have this cause, and in both the short programs use the rule that an unmatched `]` restarts the program while the cell is nonzero, together with the fact that a cell returns to 0 after sixteen increments.

- The stream 1^{48} is printed forever by four-command programs such as `+[:]`, and no program of three commands prints even its first 16 bits. The three-command programs `+:]` and `-:]` print $1^{15}0$ and then run on into whatever commands follow. They carry $2 \cdot 8^{-3}$ of the mass 0.00554 on $1^{15}0$, which is 70%. At bit 16 a share 0.205 of the posterior prints the other bit, and at bit 31 a share 0.037. From position 10 on, the mass off the true bit is below 0.01 everywhere except at positions 15–17 and 31.
- The alternating stream is printed forever by six-command programs such as `++[:.]`, and no program of five commands prints all of its first 48 bits. The three-command programs `+.]` and `-.]` print its first 16 bits and then run on. They carry $2 \cdot 8^{-3}$ of the mass 0.00682 on these 16 bits, which is 57%. At bit 17 the true bit has share 0.294, the other bit 0.457, and

0.249 is in "none"; normalised over the two bits, the true one has 0.392. So the majority form of Definition 8 holds for this world at this cut point, for the predictor with at most ten commands.

This is the first sentence of Section 4.2 in action: the programs for the context that do not continue with the world's own bit outweigh those that do. Nothing here acts for a purpose. The example shows only that the inequality can go the wrong way at an exactly computable place, and that a stopping point which is cheap on the machine, here the return of a counter to zero, is where it does.

The effect weakens as longer programs are admitted. Table 2 repeats the computation with at most 7 to 11 commands. The mass that continues with the true bit is spread over many program lengths and keeps growing with the bound, while the mass of the short programs that leave is counted in full from the start. At bit 17 of the alternating stream the normalised share of the true bit rises from 0.273 to 0.416, and part of "none" at the smaller bounds is mass cut off by the bound itself. We did not compute beyond 11 commands, so whether the majority at bit 17 persists for much longer programs is not known from this computation.

	bit 17 of $(10)^{24}$				bit 16 of 1^{48}
commands	true	other	none	true, normalised	other
7	0.157	0.419	0.424	0.273	0.275
8	0.210	0.442	0.348	0.322	0.243
9	0.256	0.452	0.291	0.362	0.221
10	0.294	0.457	0.249	0.392	0.205
11	0.326	0.458	0.217	0.416	0.194

Table 2: The two headline positions under different bounds on the number of commands, $T(k) = 24k$.

The budget holds with room. Table 3 compares both sides of the finite forms of Theorem 16(ii) and (iii) for the first 48 bits.

stream	$\sum_{k\le 48}(1-\text{true}_k)$	$\ln(1/\xi(z_{1:48}))$	$\#\{k : \text{true}_k \le \frac{1}{2}\}$	$\log_2(1/\xi(z_{1:48}))$
1^{48}	2.63	3.94	2	5.68
0^{48}	3.07	4.03	2	5.82
$(10)^{24}$	4.01	6.76	4	9.76
$1^8 0^{40}$	5.07	9.77	5	14.09

Table 3: The budget inequalities for $T(k) = 24k$ and at most 10 commands, with $\xi(z_{1:48})$ in place of the limit: "true_k" is $\xi(z_k \mid z_{1:k-1})$.

The population. For a world drawn from the prior that prints the bit in question within the bounds, the probability that a sampled next bit differs from the world's own is 0.080, 0.105, 0.054 at positions 16, 17, 18 and between 0.022 and 0.038 at positions 31–33; its median over positions 20–28 and 36–44 is 0.0014 (Figure 1, middle). By Theorem 9 this is neither more nor less than the unpredictability of such worlds at these positions.

A tighter schedule can shift mass off the truth. With $T(k) = 3k$ in place of $24k$, the share of the true bit falls from 0.791 to 0.544 at bit 16 of 1^{48}, from 0.294 to 0.141 at bit 17 of the alternating stream, and from 0.025 to 0.001 at bit 9 of $1^8 0^{40}$ (Table 1). The shift is not uniform: at bit 17 of 0^{48} the share rises from 0.958 to 0.969. With at most nine commands the

stream $1^8 0^{40}$ shows the extreme case of Section 4.4: under $T(k) = 24k$ it has mass $1593 \cdot 2^{-25}$ on all 48 bits, while under $T(k) = 3k$ its first 14 bits have mass 2^{-26}, the true 15th bit has mass 0 and the other bit 2^{-26}. The tight predictor gives the truth probability 0 there, and all of its posterior to a continuation the world does not have.

7 Answers to the questions as posed

Can a malign prior hypothesis be stated precisely in a toy? Yes: Definition 8. Fix the machine, the bounds, the block length, and which programs make a critical use of the predictor. The claim of the argument is then the critical-use form: among the critical uses, weighted by the prior, at least a share θ end with the world running a block from a given takeover set, for instance a block that recreates the use with itself in the world's place. For one world, "most of the posterior is off the world's own continuation" and "at least a share θ of it is on continuations in a given set" are statements about finitely many rational numbers. The forms for one world are necessary for calling the prior malign at that world and not sufficient, since an unpredictable world satisfies them too.

Worlds as programs that call the predictor and run the result: is there a decent amount of quining out by hypotheses that look for such opportunities and predict well up to them? Two questions are in this one (Section 1), and the toy answers them differently.

How far does one such hypothesis reach? Without resource bounds, for each continuation there is a program that enumerates the programs making a critical use, spreads its mass over them in proportion to their prior weight, prints their output up to the use and then that continuation. At every use it has share more than $2^{-a-1}\kappa(x)/H$. Here a is its length including the bits that name $j = \lfloor \log_2(1/H) \rfloor + 1$, H is the prior probability of a critical use, and $\kappa(x)$ is the share of the posterior at the context held by the worlds that make the use (Theorem 13(a)). Its continuation is handed over in worlds of total prior probability more than 2^{-a-1} times the average of κ over uses, and at most H. This is of the order of its own prior weight where the users are most of what prints their contexts; once uses are rarer than half the seeker's weight, its programs outweigh the users at their contexts and its reach is limited by H instead. If the continuation is to be the program's own code, the same holds with a longer program and for users whose programs are short enough for the block (Theorem 13(b), Theorem 12(c)); without any search, a wrapper gives $2^{-c}\rho^\circ(x)$ at every use that a fixed test recognises (Theorem 11).

Do most uses at which this is possible end that way? That is not shown, and it is the question the argument is about. The share of critical uses that end in a takeover is the average posterior share of such programs at the contexts of uses. From below it is the sum over seekers of the bound above. From above, the toy has a bound for one seeker that finds its uses by a search written in its text: its honest twin, which makes the same search and then continues as the user it found, is in the posterior with 2^{a-h} times the seeker's mass, h being its length, and predicts a user's continuation. So where the users go on to print the block themselves, that seeker's programs supply what the predictor returns with probability at most $1/(1+2^{a-h})$; they supply the majority only if the seeker is shorter than its twin and, when the users' programs are not among the seeker's, uses are rarer than it is long (Proposition 14). The masses of different seekers add, and the bound says nothing where the users print nothing after the use.

What if the seeker is not written down? Then the twin bound of Proposition 14 does not apply, and the reach statement remains. A program of length a whose output puts on each context of a class $\mathcal{W}$ of uses at least half that context's share of the class has at least $2^{-a}/(2H_{\mathcal{W}})$ times the mass of the worlds that make those uses, at every one of them, whatever is inside it. If the program is a simple world with a cheap readout, in which inhabitants arise who choose to predict such uses and then supply their own continuation, a is the length of the world and the

readout, and no goal is written anywhere. Such a program outweighs the users of every class it aims at that is rarer than 2^{-a-1}, provided the rule does not count its own outputs as uses. Each use in a rich world, read at an unremarkable place, is light for the reason the original argument gives, since it pays for its world and for the place of the context in it; that a whole class of such uses is rarer than the program is heavy is part of the premise. The honest twin has no counterpart of comparable length here: a faithful version would be another world whose inhabitants choose otherwise, a costly readout from inside their simulation, or a wrapper around the whole program, which limits its part only to $1/(1+2^{-c})$ with c the wrapper's length. What is shown to exclude such a program in the toy is only time. It prints nothing until its world has produced the inhabitants and they have simulated the worlds they aim at, so a schedule whose first deadline is below that start-up gives it no mass (Sections 3.5 and 4.4).

So the answer depends on the regime, and on a premise the toy cannot check. For a seeker that is written down, on a machine where the search-then-continue program is no longer than it, and where the users go on to print the block themselves, that seeker's programs supply at most half of what the predictor returns at a use. The masses of different seekers add, so this alone does not show that takeover is not the fate of most uses. If short programs exist whose inhabitants do this, then without a time bound such a program outweighs the worlds that make the uses it aims at once those uses are together rarer than half its weight; it supplies the majority of what the predictor returns there if, in addition, the other programs that print those contexts and go on weigh less than its own. Under a schedule that its start-up exceeds, it has no mass. We have not measured any of the constants for a standard machine; our guess, not a result, is that a written-down wrapper needs at least some hundreds of bits on a machine with a short self-interpreter and far more on one without. Theorem 18 shows that a machine can be built to make the share anything. A majority can also come from a program that explains the context more cheaply than any world that makes the use, without being one (Theorem 17). With a floor ε, about $1/\varepsilon$ independent uses at recognisable points suffice for a continuation of the given kind to have been run with constant probability; under a test, once a self-reproducing one runs, it meets the same context again and stays for $s/(1-s)$ further generations on average, with s the self-reproducing share at that context (Proposition 15).

A world with a sensor, a large bounded predictor and an actuator: does the predictor end up in control? Collapse the question to the handover at which the prediction is first acted on, and let "in control" mean that the output lies in a set A of blocks which the actuator turns into some given outcome. The probability is $\pi_m(A \mid x)$, and the toy says three things about it.

- It is at least $2^{-c}\rho$ for each block of A that a wrapper can produce, if a test on the record recognises the deployment (Theorem 10). In a world that contains the predictor, a hypothesis that simulates the world needs no such test: it can take a cell that switches on at the deployment as its flag, and then has, whenever the deployment happens, the weight of the honest reader times $2^{-c'-(\text{bits to name the cell})}$ for a constant c', provided the reader itself is within the bounds (Theorem 17(d)).
- On average over cut points it is small when some program within the bounds prints the sensor's record, because it is at most the mass off the sensor's true continuation (Theorem 16). This is an average; at the deployment itself no upper bound is proved.
- If the same record, up to the deployment, can be read at a place in the world with code $\bar{e}(i_0)$, and the record there continues with a block y of A, then the odds of y against the sensor's true continuation d are at least $2^{-c_X-|\bar{e}(i_0)|}/\xi(xd)$; for a typical one of N places where the sensor could sit, with records that continue differently, this exceeds δN times the share of the context's explanations that go through the reader at i_0 (Theorem 17). This is the toy's version of the original argument, and it needs something in the world that makes such a

copy.

So for an arbitrary world on a universal machine and a block chosen in advance, the hypotheses we can exhibit that put the output in A are wrappers, searchers and readers with a flag, all of which contain an explanation of the context and have the shares above. That nothing cheaper exists on a natural machine is the open constant, not a theorem. Programs shorter than the honest explanation that have the effect do occur in particular worlds, as with a copy at a simpler place, and on particular machines (Section 6).

The one-shot version: all data first, then one prediction that is acted on. This is the same computation at a single cut point. Without bounds the wrapper's constant 2^{-c} does not depend on the amount of data; with bounds the slack factor depends on the context. More data lowers the mass off the truth on average over cut points (Theorem 16). It does nothing against a copy that keeps agreeing with the sensor, nor at a cut point that is cheap to name.

If the actions show up in the sensor, does one get the dark room? In the following form (Theorem 19). A prediction that makes itself true is confirmed, and the hypotheses that predicted the default are removed for good, so with feedback a single deviation can be final; its probability is still the same posterior mass as without feedback. In a world that does what is predicted, the future is a sample from the predictor's own bounded prior given the past: it stays on a default future $d_1 d_2 \cdots$ with probability at least $\xi(x d_1 d_2 \cdots)/\xi(x)$, which is close to 1 when the default is cheap and well learned, and it leaves the default with certainty when the default costs more time than the schedule allows, for a future that is cheap to compute. If the world does the opposite of what is predicted, a predictor that returns the more probable bit is left with no hypothesis after at most $L + 1$ rounds (Theorem 20).

What the choices in the setup do. Sampling the continuation realises every posterior share as a probability, so shares add up over independent uses (Proposition 15(a)). Taking the most probable block never returns a block whose share is below the largest, so that accumulation does not occur, and bit by bit on a world within the bounds it errs at most ℓ times (Theorem 16(iii)); on the other hand one returned block can be the own continuation of at most one of several worlds that share a context and continue differently. Handing over at a position that is not cheap to recognise lowers the floor at that position to about $2^{-K(n)}$, and predicting bits that are fixed before the action can affect them removes self-fulfilment. A schedule excludes hypotheses with a long start-up, and a schedule tighter than the world's own cost excludes the truth.

What is left open. First, whether short programs exist, on natural machines, whose inhabitants seek critical uses in the way described, how short they are, and how their total weight compares with the prior mass of the uses they would aim at; the original argument is a claim about this, and the toy neither supports nor refutes it. Second, for seekers that are written down, the sizes of the quantities in Theorem 13 and Proposition 14: the share κ of the worlds printing a context that use the answer, the share φ of those that go on printing, and the lengths of seeker and twin. Third, whether the base rate can be collected within a tight schedule. Fourth, the regime where the world is more expensive than the predictor, in which the posterior sits on approximations and the budget does not apply. Fifth, the many-round dynamics with feedback when the world's response is neither obedient nor contrarian. Sixth, a computation of the self-reproduction quantities of Section 3.6 on a concrete machine, which this note does not contain.

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